



AIM: Perform steep turns at 45° and 60° AoB. Identify and recover from a spiral dive. Demonstrate slipping turns and dutch roll reversals.

P E8: Independent takeoff; no prompting at any stage ☐

DP Dutch roll at 20° AoB: Smooth coordinated rolls alternating left and right; remind student of parallax ☐

P HASSELL: Student completes – **mandatory before steep turns begin, every 15 mins** ☐

DP 45° steep turn L and R: Bank, Balance, Backpressure; overbanking tendency; hold AoB $\pm 5^\circ$, alt ± 100 ft; rollout $\pm 10^\circ$ ☐

DP 60° steep turn L and R: Same sequence; increased backpressure and overbanking; demonstrate visual difference from 45° ☐

DP Spiral dive: Instructor initiates by increasing bank and lowering nose; student practices recovery – **throttle, wings level, ease back** ☐

DP Slipping turn: Patter: decreased turn rate, opposite rudder, no temptation to pull back, ball low, pro-recovery tendency, outer wing stall risk; recover to coordinated ☐

D Skidding turn: Instructor demonstration ONLY – no student practice; Patter: increased turn rate, same rudder, yaws toward ground (temptation to pull back), ball high, pro-spin tendency; recover to coordinated ☐

COMMON ERRORS

- **Altitude loss in steep turn** -> Check bank first; if correct, adjust backpressure; reference the horizon
- **Backpressure before wings level in spiral recovery** -> **Level wings first** – non-negotiable; NC if incorrect
- **HASSELL omitted** -> NC for entire sequence

STANDARDS FOR PROGRESSION

P S **Progressing (P):** 45° $\pm 10^\circ$ bank, ± 200 ft; spiral recovery with minor prompting
Solo (S): 60° $\pm 5^\circ$ bank, ± 150 ft, rollout $\pm 10^\circ$; spiral recovery correct without prompting; HASSELL without prompting
C NC
Note: Two-flight turning requirement (E6 + E9) now satisfied. Full C requires Phase 3.

⚠ SAFETY Smooth air required · Ceiling **>2500 ft AGL** · Above 2000 ft AGL throughout · Recoveries above 1500 ft AGL · Spiral dive: instructor briefs recovery sequence before flight; take control immediately if uncontrolled · Skidding turn: instructor demo ONLY